

From Oracle to Auditor

Machine-Checked Reasoning about Gödel's God and the Impossible Devil

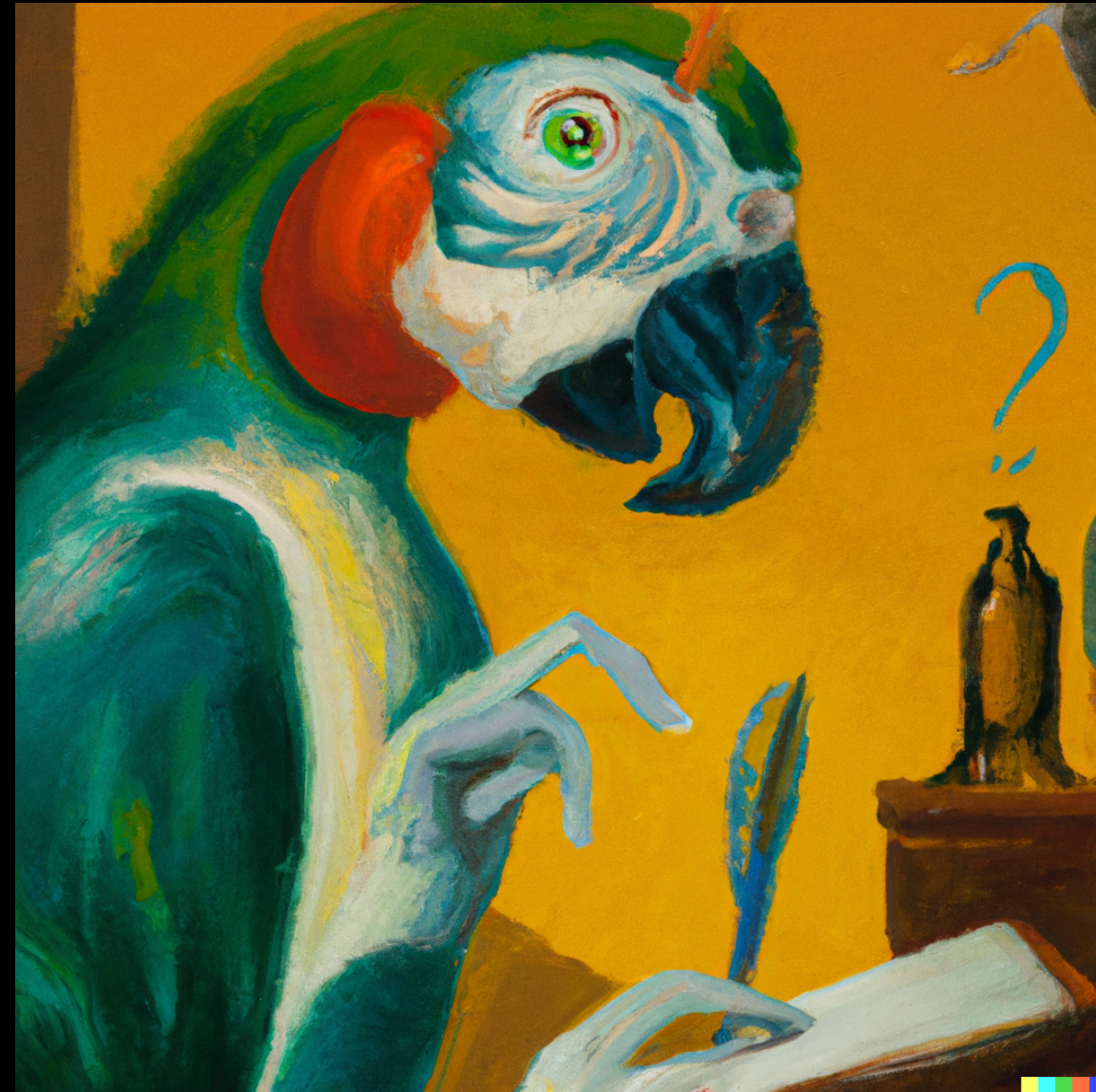
Christoph Benz Müller

U Bamberg | FU Berlin

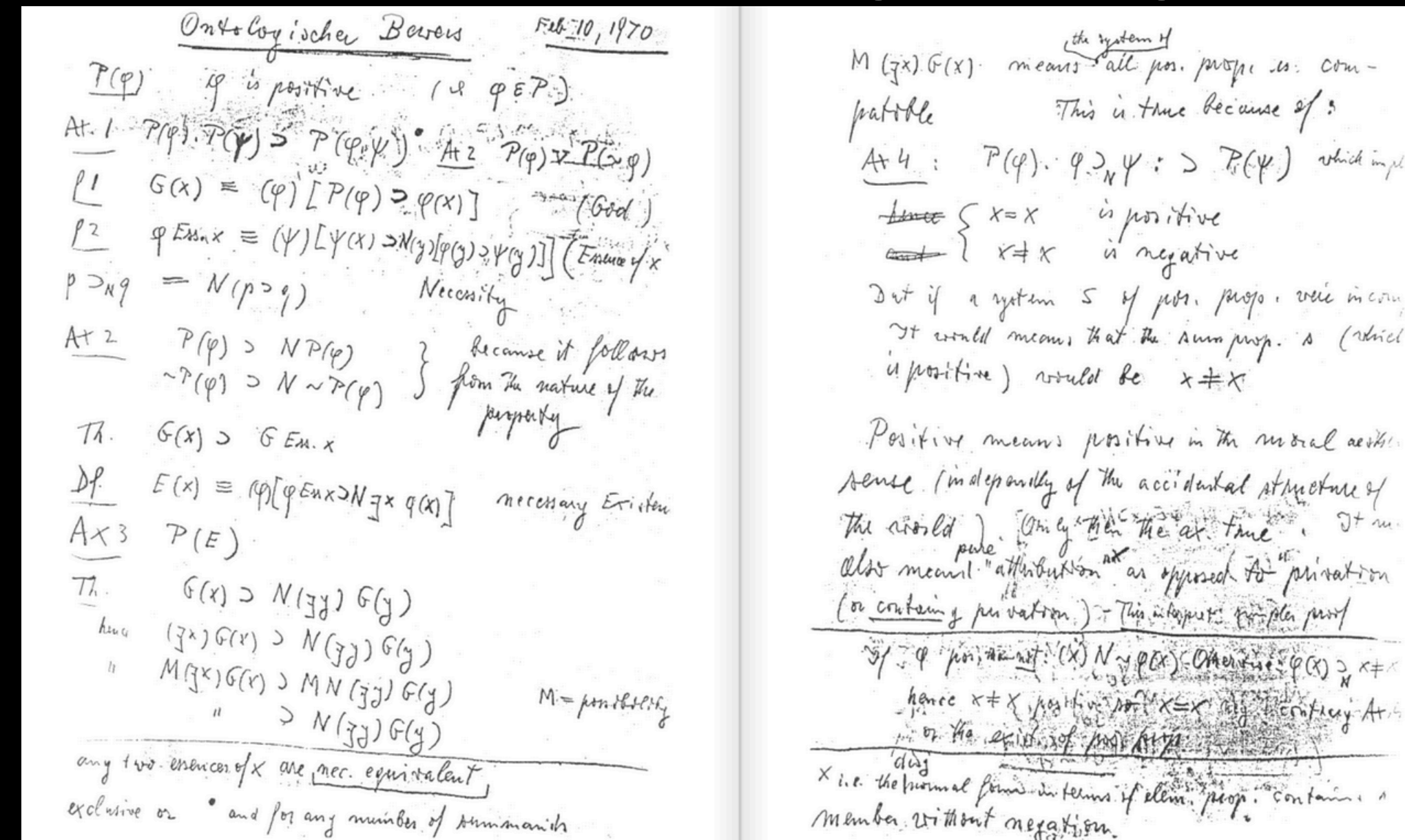
5th World Congress on Logic and Religion — Vancouver 2026

Sunday, 27 September 2026

Oracle



Gödel's Modal Ontological Argument



Auditor



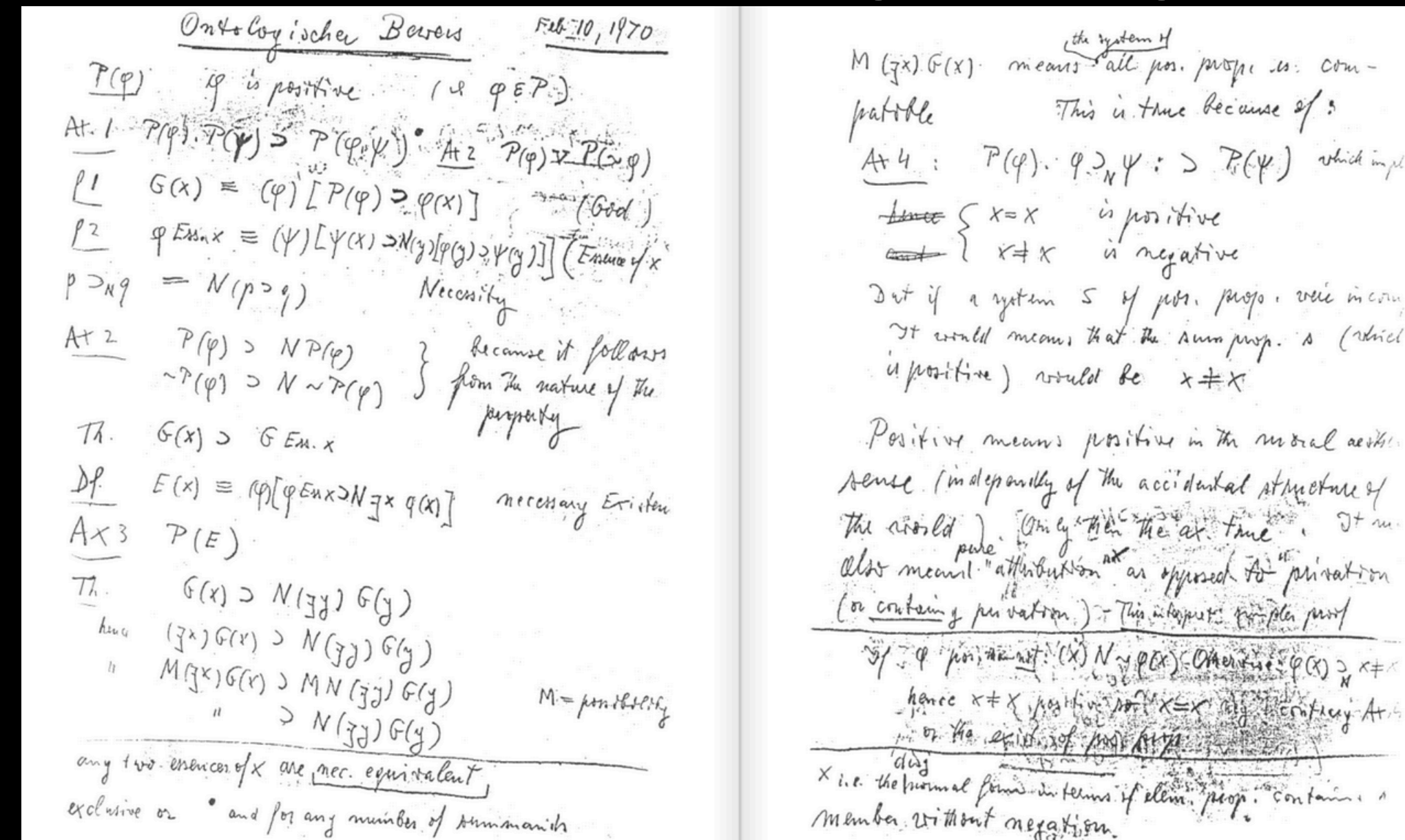
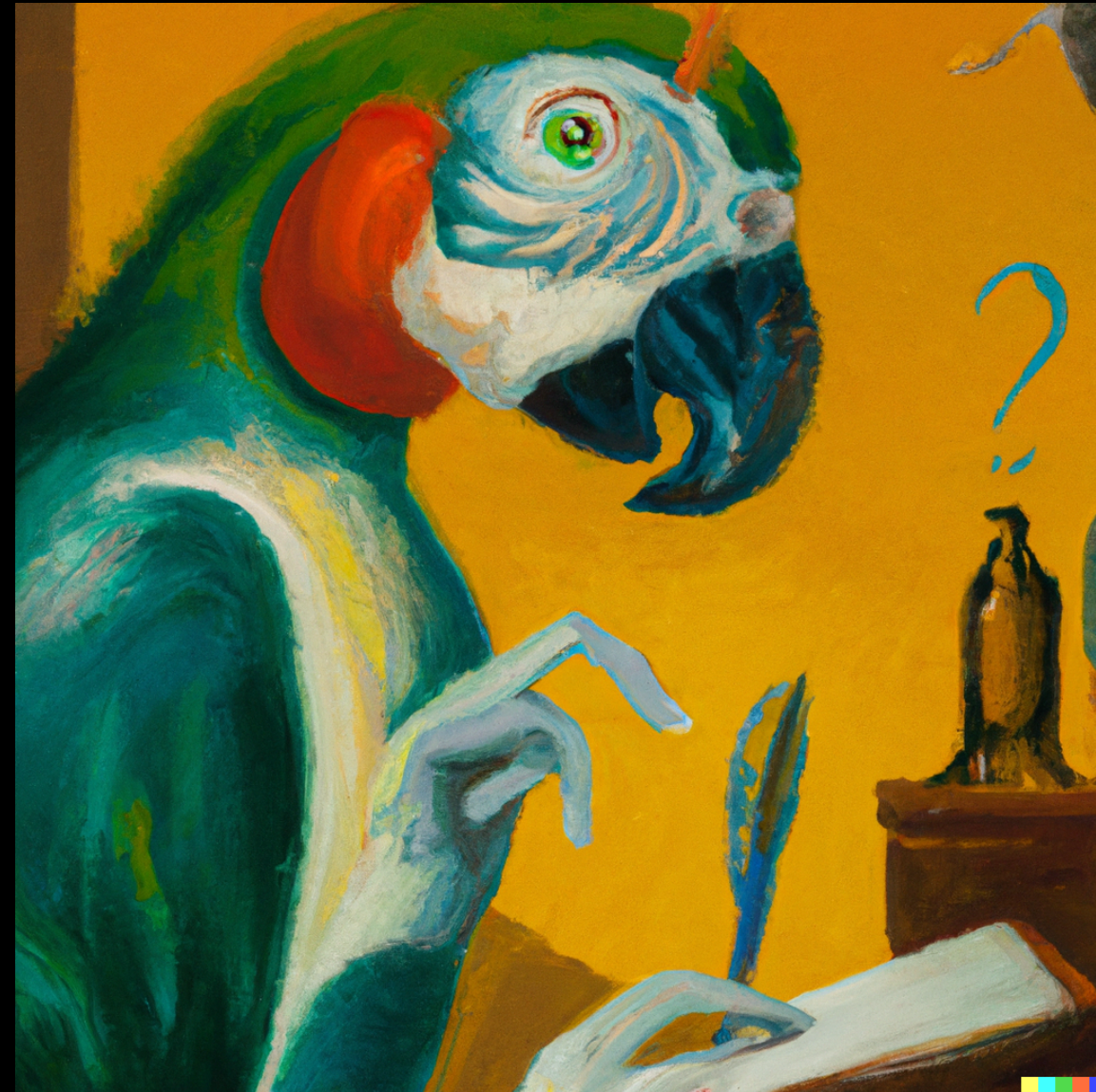
- AI Oracle: generate argument
 - fluent
 - confident
 - but can we trust?

- AI Auditor: verify argument
 - from premises, within chosen logic
 - complete, machine-checked proof
 - trust, relative to premise and logic

Oracle

Auditor

Gödel's Modal Ontological Argument

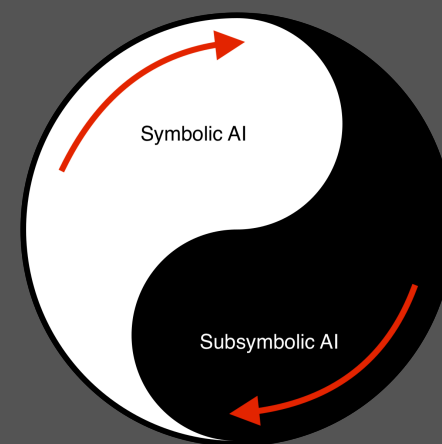


- Oracles: Claude, Codex, ...
 - draft formalisations
 - explain steps
- Auditors: Lean, Isabelle, Rocq, ...
 - verify rigorously
 - but in which logic(s): LogiKEy
- Both combined: trustworthy AI [AIJ, 2020]

Instead of Maths (in current headlines):

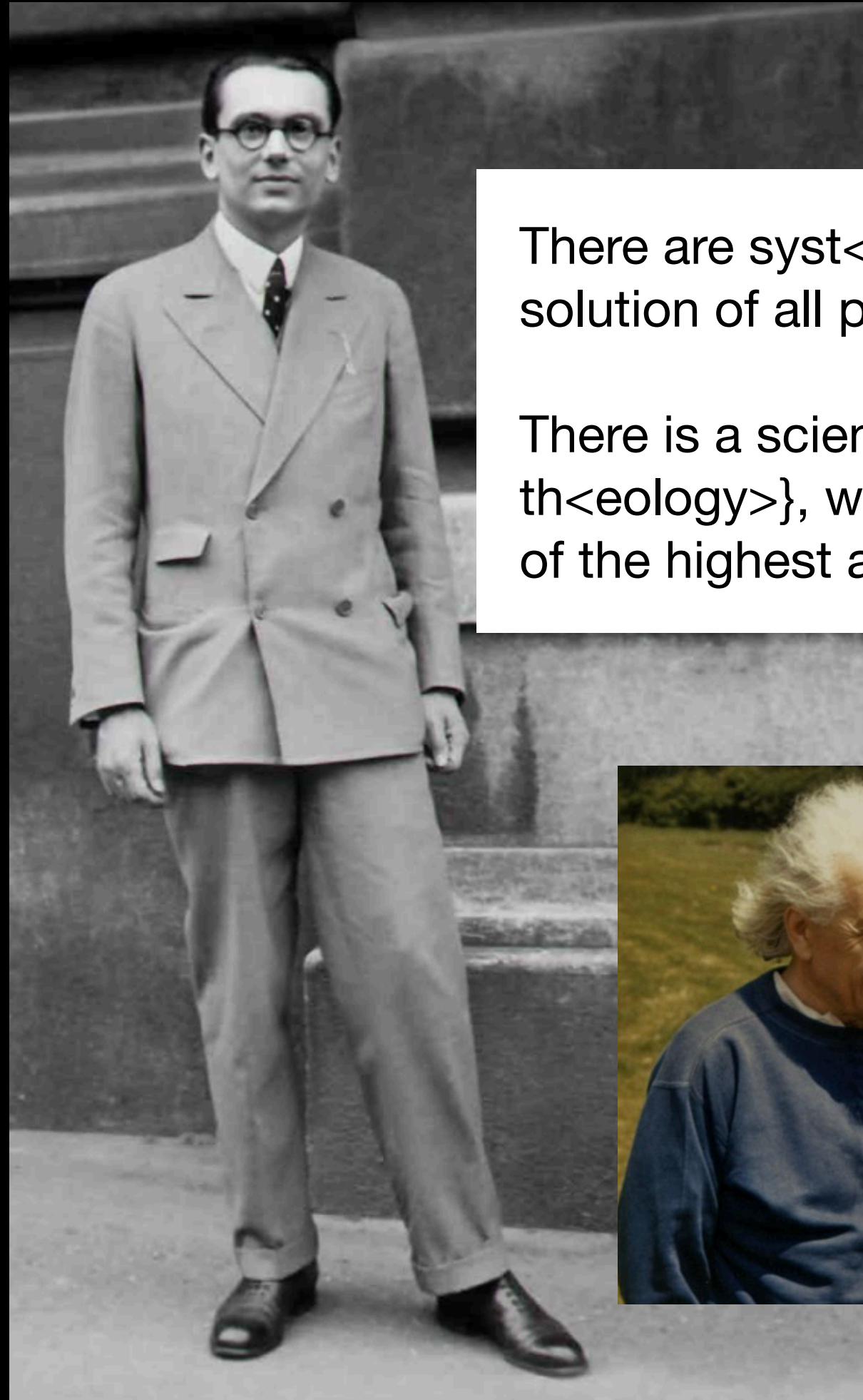
Computational Metaphysics

Formal methods for the analysis of philosophical arguments



Analysis of philosophical arguments using formal methods

'Exact' methods are possible ...



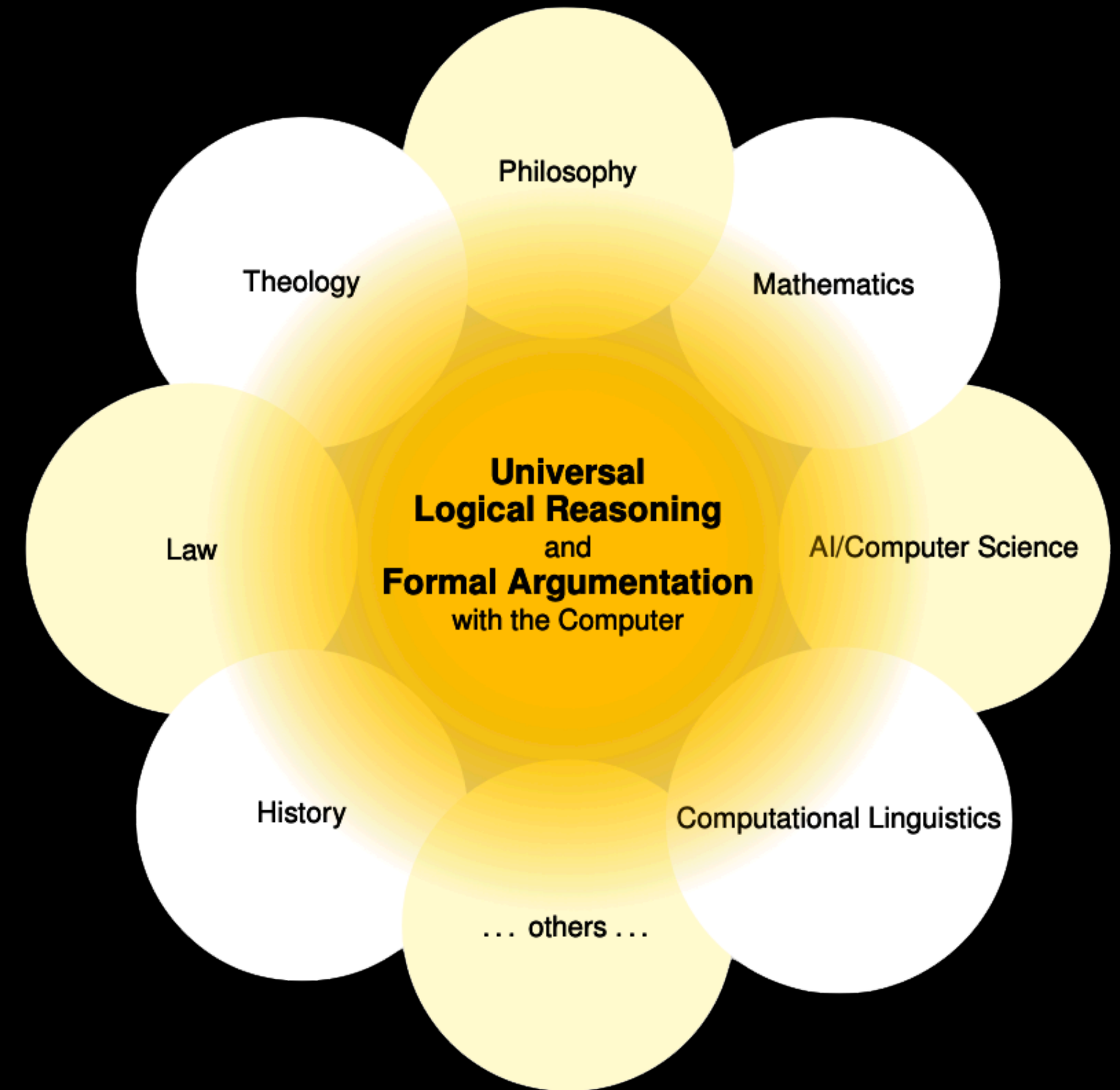
There are systematic methods for the solution of all problems (also art etc.).

There is a scientific (exact) philosophy {and theology}, which deals with the concepts of the highest abstractness.

Kurt Gödel: Maximen V



Kurt Gödel (1906-78)



... far beyond the realm of mathematics!

Rest of this presentation is based on ...

Automating Gödel's Ontological Proof of God's Existence with Higher-order Automated Theorem Provers

Christoph Benz Müller, Bruno Woltzenlogel Paleo · *ECAI 2014* pp. 93 – 98

2014

The Inconsistency in Gödel's Ontological Argument: A Success Story for AI in Metaphysics

Christoph Benz Müller, Bruno Woltzenlogel Paleo · *IJCAI 2016* pp. 936-942

2016

arXiv

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[Submitted on 26 May 2026]

Many Logics, One Methodology: A Plea for Logical Pluralism in Formalised Reasoning (preprint)

Christoph Benz Müller, Daniel Kirchner, Luca Pasetto

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Gödel's and Scott's Variants of the Ontological Argument in Lean 4

Christoph Benz Müller

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Notes on Gödel's and Scott's variants of the ontological argument

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Volume 208, pages 569–611 (2025) [Cite this article](#)

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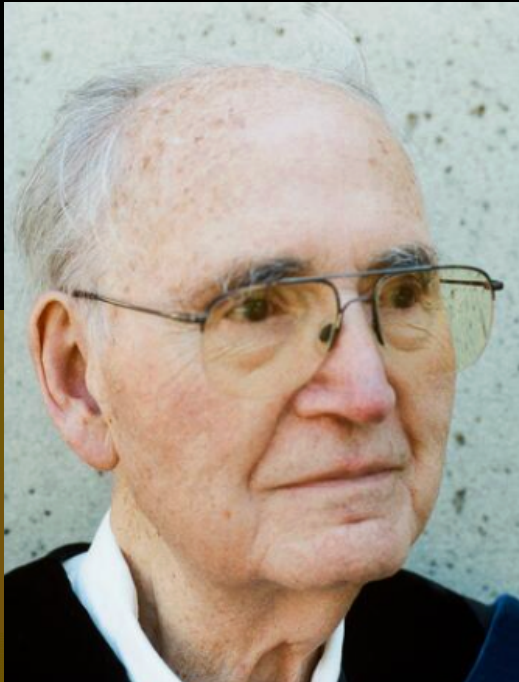
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Christoph Benz Müller & Dana Scott

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Abstract

Notes on Kurt Gödel's modal ontological argument and Dana Scott's variant of it are presented. These remarks, supported by experimental studies with a proof assistant system for classical higher-order logic, implicitly answer some questions the authors have received over the last decade(s). In addition, some new insights resulting from the conducted experiments are reported.



jww: Dana Scott

... and some further, novel insights

Ontologischer Beweis

Feb 10, 1970

$P(\varphi)$ φ is positive ($\varphi \in P$)

At 1 $P(\varphi) \cdot P(\psi) \supset P(\varphi \cdot \psi)$ At 2 $P(\varphi) \vee P(\sim \varphi)$

P1 $G(x) \equiv (\varphi) [P(\varphi) \supset \varphi(x)]$ (God)

P2 $\varphi \text{ Ess } x \equiv (\psi) [\psi(x) \supset N(y) [\varphi(y) \supset \psi(y)]]$ (Essence of x)

$p \supset_N q = N(p \supset q)$ Necessity

At 2 $P(\varphi) \supset NP(\varphi)$
 $\sim P(\varphi) \supset N \sim P(\varphi)$ } because it follows from the nature of the property

Th. $G(x) \supset G \text{ Ess } x$

Df. $E(x) \equiv (\varphi) [\varphi \text{ Ess } x \supset N \exists x \varphi(x)]$ necessary Existence

Ax 3 $P(E)$

Th. $G(x) \supset N(\exists y) G(y)$

hence $(\exists x) G(x) \supset N(\exists y) G(y)$

" $M(\exists x) G(x) \supset MN(\exists y) G(y)$
 " $\supset N(\exists y) G(y)$ M = possibility

any two essences of x are nec. equivalent

exclusive or and for any number of summands

higher-order

modal

important!

$M(\exists x) G(x)$ means ^{the system of} all pos. prop. is compatible
 This is true because of:

At 4: $P(\varphi) \cdot \varphi \supset_N \psi \supset P(\psi)$ which impl

~~hence~~ $\begin{cases} x=x & \text{is positive} \\ x \neq x & \text{is negative} \end{cases}$

But if a system S of pos. prop. were incomp.
 It would mean that the sum prop. S (which is positive) would be $x \neq x$

Positive means positive in the moral aesthetic sense (independently of the accidental structure of the world). Only then the ax. true. It also means "attribution" as opposed to "privation" (or containing privation). This interprets simpler proof

If φ pos. then $(x) N \sim \varphi(x)$ Otherwise $\varphi(x) \supset_N x \neq x$

hence $x \neq x$ positive so $x=x$ neg. contradictory At

of the exist. of pos. prop.

x i.e. the normal form in terms of elem. prop. contains a member without negation.

Overview

- HOML in HOL
- ~~Scott's variant (with PG, as we already know it)~~
- Gödel's scriptum variant (1970)
 - a. **Gödel1**: scriptum notion of essence — inconsistency
 - b. **Gödel2**: modified notion of essence — $\Box \exists G$
 - c. **Gödel3**: modified notion of property inclusion — $\Box \exists G$
 - d. **No (perfect) Devil (or Devil = God)** — $\Box \exists G$
- Gödel's scriptum variant (1970) and mathematical realism is (uncountably/maximally) infinite
- Personal conclusion

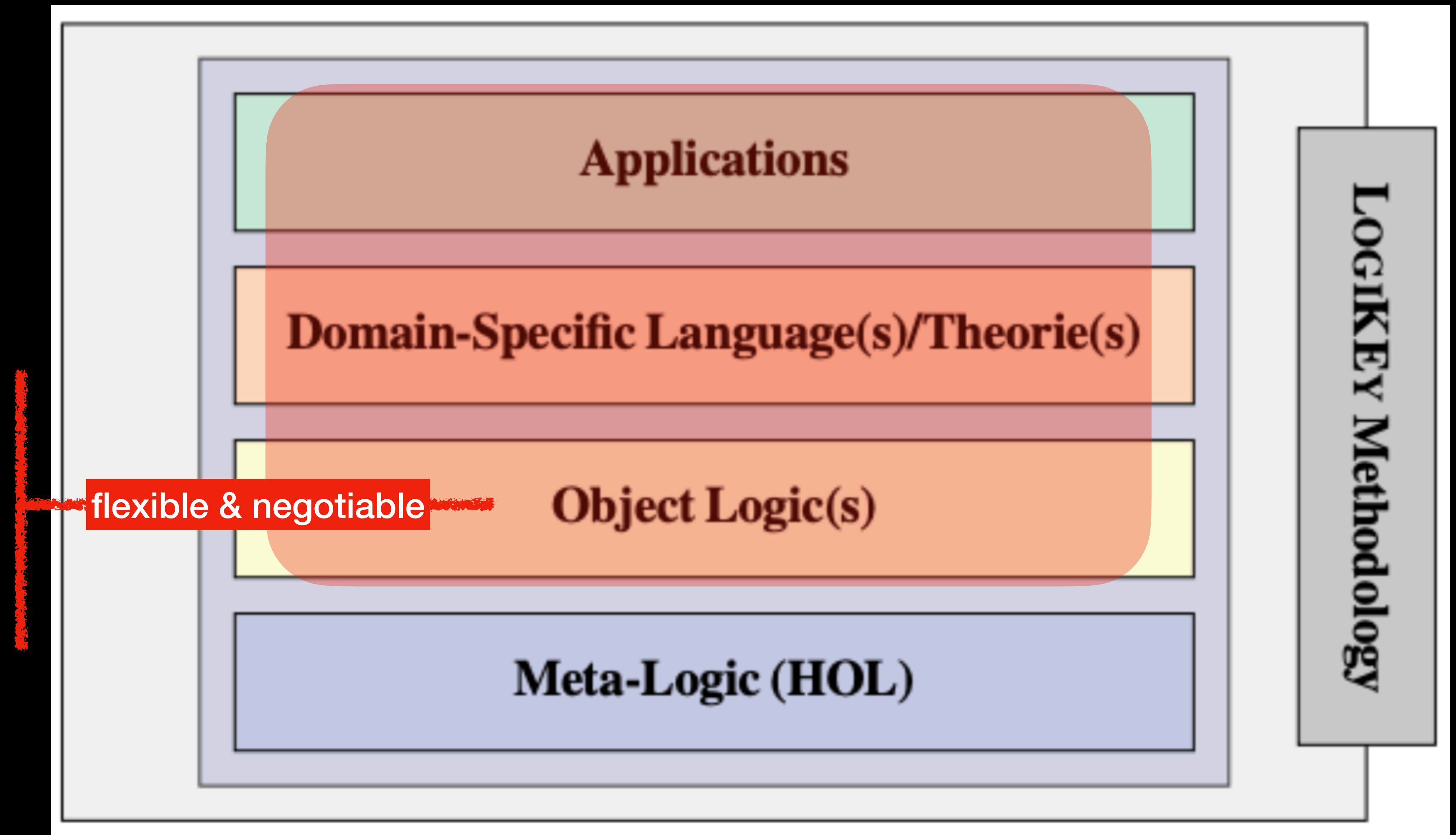
new formal studies

new variant

— (

Logic-Pluralistic KR & R: LogiKEy

Different modal logics:
K, T, KB, S4, S5, etc
Different quantifiers
Extensional or intensional
etc.



Origins: 2007-2008 (with L. Paulson)

Term LogiKEy: 2017-2019 (with L. van der Torre & X. Parent)

LogiKEY@work

HOML in HOL

HOML in HOL



A mathematician would only look at these equations

```

1 theory HOMLinHOL imports Main
2 begin
3 —<Global parameters setting for the model finder nitpick and the parser; unimport for the reader>
4 nitpick_params[user_axioms,expect=genuine,show_all,format=2,max_genuine=3]
5 declare[[syntax_ambiguity_warning=false]]
6 —<Type i is associated with possible worlds and type e with entities:>
7 typedecl i —<Possible worlds>
8 typedecl e —<Individuals/entities>
9 type_synonym  $\sigma$  = "i $\Rightarrow$ bool" —<World-lifted propositions>
10 type_synonym  $\tau$  = "e $\Rightarrow$  $\sigma$ " —<Modal properties>
11 consts R::"i $\Rightarrow$ i $\Rightarrow$ bool" ("_r_") —<Accessibility relation between worlds>
12 axiomatization where
13   Rrefl: " $\forall x. xrx$ " and Rsymm: " $\forall x y. xry \longrightarrow yrx$ " and Rtrans: " $\forall x y z. xry \wedge yrz \longrightarrow xrz$ "
14 —<Logical connectives (operating on truth-sets):>
15 abbreviation Mbot:: $\sigma$  ("⊥") where
16 abbreviation Mtop:: $\sigma$  ("⊤") where
17 abbreviation Mneg::" $\sigma \Rightarrow \sigma$ " ("¬_") [52]53) where
18 abbreviation Mand::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∧" 50) where
19 abbreviation Mor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨" 49) where
20 abbreviation Mimp::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "⊃" 48) where
21 abbreviation Mequiv::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "↔" 47) where
22 abbreviation Mbox::" $\sigma \Rightarrow \sigma$ " ("□_") [54]55) where
23 abbreviation Mdia::" $\sigma \Rightarrow \sigma$ " ("◇_") [54]55) where
24 abbreviation Mprimeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("=_") where
25 abbreviation Mprimneg::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≠_") where
26 abbreviation Mnegpred::" $\tau \Rightarrow \tau$ " ("¬~_") where
27 abbreviation Mconpred::" $\tau \Rightarrow \tau \Rightarrow \tau$ " (infixl "." 50) where
28 abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨e" 49) where
29 —<Possibilist quantifiers (polymorphic)>
30 abbreviation Mallposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀") where
31 abbreviation Mallpossb (binder "∀" [8]9) where
32 abbreviation Mexiposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃") where
33 abbreviation Mexipossb (binder "∃" [8]9) where
34 —<Actualist quantifiers (for individuals/entities)>
35 consts existsAt::"e $\Rightarrow$  $\sigma$ " ("_@_")
36 abbreviation Mallact::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("∀E") where
37 abbreviation Mallactb (binder "∀E" [8]9) where
38 abbreviation Mexiact::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("∃E") where
39 abbreviation Mexiactb (binder "∃E" [8]9) where
40 —<Leibniz equality (polymorphic):>
41 abbreviation Mleibeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≡_") where
42 —<Meta-logical predicate for global validity:>
43 abbreviation Mvalid::" $\sigma \Rightarrow$ bool" ("⊢_") where
44 end

```

```

"⊥ ≡ λw. False"
"⊤ ≡ λw. True"
"¬φ ≡ λw. ¬(φ w)"
"φ ∧ ψ ≡ λw. φ w ∧ ψ w"
"φ ∨ ψ ≡ λw. φ w ∨ ψ w"
"φ ⊃ ψ ≡ λw. φ w ⟶ ψ w"
"φ ↔ ψ ≡ λw. φ w ⟷ ψ w"
"□φ ≡ λw. ∀v. w r v ⟶ φ v"
"◇φ ≡ λw. ∃v. w r v ∧ φ v"
"x = y ≡ λw. x = y"
"x ≠ y ≡ λw. x ≠ y"
"¬Φ ≡ λx. λw. ¬Φ x w"
"Φ.Ψ ≡ λx. λw. Φ x w ∧ Ψ x w"
"φ ∨e ψ ≡ (φ ∨ ψ) ∧ ¬(φ ∧ ψ)"

"∀Φ ≡ λw. ∀x. Φ x w"
"∀x. φ(x) ≡ ∀φ"
"∃Φ ≡ λw. ∃x. Φ x w"
"∃x. φ(x) ≡ ∃φ"

"∀EΦ ≡ λw. ∀x. x@w ⟶ Φ x w"
"∀Ex. φ(x) ≡ ∀Eφ"
"∃EΦ ≡ λw. ∃x. x@w ∧ Φ x w"
"∃Ex. φ(x) ≡ ∃Eφ"

"x ≡ y ≡ ∀P. P x ⊃ P y"

"⊢ψ ≡ ∀w. ψ w"

```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKey methodology

HOML in HOL: tests

S5 axioms: valid as expected

Barcan formulas: not valid for 'varying-dom.' quant.

Barcan formulas: valid for 'constant-dom.' quant.

Hilbert axioms and rules: valid as expected

Existential import: yes, for 'constant-domain' quant.

Existential import: no, for 'varying-domain' quant.

Properties of equality: valid as expected
(except for Boolean extensionality)

Full comprehension: valid as expected

Modal collapse: not valid

universal property = self-identity

Absurd/empty property = self-difference

```

1 theory TestsHOML imports HOMLinHOL
2 begin
3 <Test for S5 modal logic>
4 lemma axM: "[⊢ φ ⊃ φ]" using Rrefl by blast
5 lemma axD: "[⊢ φ ⊃ ◇φ]" using Rrefl by blast
6 lemma axB: "[⊢ φ ⊃ □◇φ]" using Rsymm by blast
7 lemma ax4: "[⊢ φ ⊃ □□φ]" using Rtrans by blast
8 lemma ax5: "[⊢ ◇φ ⊃ □◇φ]" using Rsymm Rtrans by blast
9 <Test for Barcan and converse Barcan formula:>
10 lemma BarcanAct: "[⊢ (∀x. □(φ x)) ⊃ □(∀x. (φ x))]" nitpick oops <Countermodel found>
11 lemma ConvBarcanAct: "[⊢ □(∀x. (φ x)) ⊃ (∀x. □(φ x))]" nitpick oops <Countermodel found>
12 lemma BarcanPoss: "[⊢ (∀x. □(φ x)) ⊃ □(∀x. φ x)]" by blast
13 lemma ConvBarcanPoss: "[⊢ □(∀x. (φ x)) ⊃ (∀x. □(φ x))]" by blast
14 <A simple Hilbert system for classical propositional logic is derived>
15 lemma Hilbert_A1: "[⊢ A ⊃ (B ⊃ A)]" by blast
16 lemma Hilbert_A2: "[⊢ (A ⊃ (B ⊃ C)) ⊃ ((A ⊃ B) ⊃ (A ⊃ C))]" by blast
17 lemma Hilbert_MP: assumes "[⊢ A]" and "[⊢ A ⊃ B]" shows "[⊢ B]" using assms by blast
18 <We have a polymorphic possibilist quantifier for which existential import holds>
19 lemma Quant_1: assumes "[⊢ A]" shows "[⊢ ∀x::'a. A]" using assms by auto
20 <Existential import holds for possibilist quantifiers>
21 lemma ExImPossibilist1: "[⊢ ∃x::e. x = x]" by blast
22 lemma ExImPossibilist2: "[⊢ ∃x::e. x ≡ x]" by blast
23 lemma ExImPossibilist3: "[⊢ ∃x::e. x = t]" by blast
24 lemma ExImPossibilist4: "[⊢ ∃x::'a. x ≡ t::'a]" by blast
25 lemma ExImPossibilist: "[⊢ ∃x::'a. T]" by blast
26 <We have an actualist quantifier for individuals for which existential import does not hold>
27 lemma Quant_2: assumes "[⊢ A]" shows "[⊢ ∀x::e. A]" using assms by auto
28 <Existential import does not hold for our actualist quantifiers (for individuals)>
29 lemma ExImActualist1: "[⊢ ∃x::e. x = x]" nitpick[card=1] oops <Countermodel found>
30 lemma ExImActualist2: "[⊢ ∃x::e. x ≡ x]" nitpick[card=1] oops <Countermodel found>
31 lemma ExImActualist3: "[⊢ ∃x::e. x = t]" nitpick[card=1] oops <Countermodel found>
32 lemma ExImActualist: "[⊢ ∃x::e. T]" nitpick[card=1] oops <Countermodel found>
33 <Properties of the embedded primitive equality, which coincides with Leibniz equality>
34 lemma EqRef1: "[⊢ x = x]" by blast
35 lemma EqSym: "[⊢ (x = y) ↔ (y = x)]" by blast
36 lemma EqTrans: "[⊢ ((x = y) ∧ (y = z)) ⊃ (x = z)]" by blast
37 lemma EQCong: "[⊢ (x = y) ⊃ ((φ x) = (φ y))]" by blast
38 lemma EQFuncExt: "[⊢ (φ = ψ) ⊃ (∀x. ((φ x) = (ψ x)))]" by blast
39 lemma EQBoolExt1: "[⊢ (φ = ψ) ⊃ (φ ↔ ψ)]" by blast
40 lemma EQBoolExt2: "[⊢ (φ ↔ ψ) ⊃ (φ = ψ)]" nitpick[card=2] oops <Countermodel found>
41 lemma EQBoolExt3: "[⊢ (φ ↔ ψ) → [(φ = ψ)]]" by blast
42 lemma EqPrimLeib: "[⊢ (x = y) ↔ (x ≡ y)]" by auto
43 <Comprehension is natively supported in HOL (due to λ-abstraction)>
44 lemma Comprehension1: "[⊢ ∃φ. ∀x. (φ x) ↔ A]" by force
45 lemma Comprehension2: "[⊢ ∃φ. ∀x. (φ x) ↔ (A x)]" by force
46 lemma Comprehension3: "[⊢ ∃φ. ∀x y. (φ x y) ↔ (A x y)]" by force
47 <Modal collapse does not hold>
48 lemma ModalCollapse: "[⊢ ∀φ. φ ⊃ □φ]" nitpick[card=2] oops <Countermodel found>
49 <Empty property and self-difference>
50 lemma TruePropertyAndSelfIdentity: "[⊢ (λx::e. T) = (λx. x = x)]" by blast
51 lemma EmptyPropertyAndSelfDifference: "[⊢ (λx::e. ⊥) = (λx. x ≠ x)]" by blast
52 lemma EmptyProperty2: "[⊢ ∃x. φ x → [φ ≠ (λx::e. ⊥)]]" by blast
53 lemma EmptyProperty3: "[⊢ ∃x. φ x → [φ ≠ (λx::e. ⊥)]]" by blast
54
55 end
56

```

Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

HOML in HOL: modal (ultra)filter

Set operations
lifted to
the HOML level

```
1 theory ModalFilter imports HOMLinHOL
2 begin
3 type_synonym  $\tau = "e \Rightarrow \sigma"$ 
4 abbreviation Element:: $\tau \Rightarrow (\tau \Rightarrow \sigma) \Rightarrow \sigma$  (infix  $\in$  90) where
5 abbreviation EmptySet:: $\tau$  (" $\emptyset$ ") where
6 abbreviation UniversalSet:: $\tau$  (" $U$ ") where
7 abbreviation Subset:: $\tau \Rightarrow \tau \Rightarrow \sigma$  (infix  $\subseteq$  80) where
8 abbreviation SubsetE:: $\tau \Rightarrow \tau \Rightarrow \sigma$  (infix  $\subseteq^E$  80) where
9 abbreviation Intersection:: $\tau \Rightarrow \tau \Rightarrow \tau$  (infix  $\sqcap$  91) where
10 abbreviation Inverse:: $\tau \Rightarrow \tau$  (" $^{-1}$ ") where
11 abbreviation "Filter  $\Phi \equiv U \in \Phi \wedge \neg(\emptyset \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \varphi \subseteq^E \psi \supset \psi \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \psi \in \Phi \supset \varphi \sqcap \psi \in \Phi)"$ 
12 abbreviation "UFilter  $\Phi \equiv \text{Filter } \Phi \wedge (\forall \varphi. \varphi \in \Phi \vee (-^1 \varphi) \in \Phi)"$ 
13 abbreviation "FilterP  $\Phi \equiv U \in \Phi \wedge \neg(\emptyset \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \varphi \subseteq \psi \supset \psi \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \psi \in \Phi \supset \varphi \sqcap \psi \in \Phi)"$ 
14 abbreviation "UFilterP  $\Phi \equiv \text{FilterP } \Phi \wedge (\forall \varphi. \varphi \in \Phi \vee (-^1 \varphi) \in \Phi)"$ 
15 end
```

$$\begin{aligned} \varphi \in S &\equiv S \varphi \\ \emptyset &\equiv \lambda x. \bot \\ U &\equiv \lambda x. \top \\ \varphi \subseteq \psi &\equiv \forall x. ((\varphi x) \supset (\psi x)) \\ \varphi \subseteq^E \psi &\equiv \forall^E x. ((\varphi x) \supset (\psi x)) \\ \varphi \sqcap \psi &\equiv \lambda x. ((\varphi x) \wedge (\psi x)) \\ {}^{-1}\psi &\equiv \lambda x. \neg(\psi x) \end{aligned}$$

Fig. 5 Set filter and ultrafilter formalized for our modal logic setting

- Notions of filter and ultrafilter lifted to the HOML level (for varying-/constant-dom. quant.)
- contains the universal set, but not the empty set
 - closed under supersets and intersection
 - maximal (only for ultrafilters)

Ontologischer Beweis Feb. 10, 1970

$P(\varphi)$ φ is positive (i.e. $\varphi \in P$)

At. 1 $P(\varphi) \cdot P(\psi) \supset P(\varphi \cdot \psi)$ At. 2 $P(\varphi) \cdot \neg P(\psi) \supset P(\varphi \cdot \neg \psi)$

[1] $G(x) \equiv (\varphi) [P(\varphi) \supset \varphi(x)]$ (Good)

[2] $\varphi \text{ Ess } x \equiv (\psi) [\psi(x) \supset N(\exists y) [\varphi(y) \supset \psi(y)]]$ (Essence of x)

$p \supset_N q = N(p \supset q)$ Necessity

At. 2 $P(\varphi) \supset NP(\varphi)$
 $\neg P(\varphi) \supset N\neg P(\varphi)$ } because it follows from the nature of the property

Th. $G(x) \supset G \text{ Ess } x$

Df. $E(x) \equiv (\varphi) [\varphi \text{ Ess } x \supset N\exists x \varphi(x)]$ necessary Existence

Ax 3 $P(E)$

Th. $G(x) \supset N(\exists y) G(y)$
 hence $(\exists x) G(x) \supset N(\exists y) G(y)$
 " $M(\exists x) G(x) \supset MN(\exists y) G(y)$ $M = possibility$
 " $\supset N(\exists y) G(y)$

any two sentences of x are nec. equivalent
 exclusive or $\cdot$ and for any number of humanoids

$M(\exists x) G(x)$ means ^{the system of} all pos. props. is compatible
 This is true because of:
 At. 4: $P(\varphi) \cdot \varphi \supset_N \psi \supset P(\psi)$ which impl.
~~hence~~ $\begin{cases} x=x & \text{is positive} \\ x \neq x & \text{is negative} \end{cases}$
 But if a system S of pos. props. were inconsistent it would mean that the same prop. is (which is positive) would be $x \neq x$

Positive means positive in the moral aesthetic sense (independently of the accidental structure of the world). (Only when the ax. true. It means also moral "attribution" as opposed to "privation" (or containing privation). This interpretation provides proof
 If $\neg \varphi$ is $\neg$ privation, then $(x) N\varphi(x)$ - otherwise $\varphi(x) \supset_N x \neq x$
 hence $x \neq x$ positive, not $x=x$ neg. necessity At. 4
 or the φ is $\neg$ privation, then $(x) N\varphi(x)$ - otherwise $\varphi(x) \supset_N x \neq x$
 hence $x \neq x$ positive, not $x=x$ neg. necessity At. 4
 i.e. the normal form in terms of elem. prop. contains a member without negation.

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Formalisation of Gödel's manuscript of 1970 (scriptum notion of essence: Gödel1)

— inconsistency —

Gödel 1

Positive properties: P

Axiom Ax1: $P\varphi \wedge P\psi \supset P(\varphi . \psi)$

Axiom Ax2a: $P\varphi \vee^e P\sim\varphi$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi . \psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
```

P: Positive properties (intensions)

Ax1:

The conjunction of two positive properties is positive.

Ax2a:

Either a property or its complement is positive.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Definition of God-like: $G x \equiv \forall \varphi. P \varphi \supset \varphi x$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall \varphi. P \varphi \supset \varphi x$ "
```

G(od-like):
An entity x is God-like
iff
it possesses all positive properties.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Definition of Essence: $\varphi \text{ Ess. } x \equiv \forall \psi. \psi x \supset (\varphi \supset_N \psi)$

The subtle defect: by this definition, the empty property is an essence of every x

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi. P \varphi \supset \varphi x$ "
7 abbreviation PropertyInclusion ("_ $\supset_N$ _") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8 definition Essence ("_Ess._") where " $\varphi \text{ Ess. } x \equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
```

Ess(ence):

A property φ is an essence of x
iff

all properties of x are necessarily implied by φ .

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Axiom Ax2b: $P \varphi \supset \Box P \varphi$ and $\neg P \varphi \supset \Box(\neg P \varphi)$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
```

Ax2b:

Positive properties are necessarily positive.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Axiom Ax2b: $P \varphi \supset \Box P \varphi$ and $\neg P \varphi \supset \Box(\neg P \varphi)$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8 definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
```

Lemma:
Non-positive properties are necessarily non-positive.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Theorem Th1: $G x \supset G \text{Ess. } x$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9   axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10  lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
```

Theorem 1:
Being God-like is an essence
of every God-like entity.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Definition of Necessary Existence: $E x \equiv \forall \varphi. \varphi \text{ Ess. } x \supset \Box(\exists^E x. \varphi x)$

Axiom Ax3: $P E$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall \varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi \text{ Ess. } x \equiv \forall \psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall \varphi. \varphi \text{ Ess. } x \supset \Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
```

E(xistence):

x has the property of existence
iff

all essences of x are necessarily exemplified.

Ax3:

E(xistence) is a positive property.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Theorem Th2: $G x \supset \Box(\exists^E y. G y)$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi \supset$  P ( $\varphi . \psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi \vee^e$  P  $\sim \varphi$ ]"
6   definition God ("G") where "G x  $\equiv \forall \varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \forall \psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall \varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset \Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
```

Theorem 2:

If x is God-like, then the property of being God-like is necessarily exemplified.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Theorem Th3: $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$

Logic KB (Rsymm) needed here

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi \wedge P \psi \supset P (\varphi \cdot \psi)$ ]"
5   axiomatization where Ax2a: "[P  $\varphi \vee^e P \sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv \forall\varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall\varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset \Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
```

Theorem 3:

The possible existence of a God-like entity implies the necessary existence of a God-like entity.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Axiom Ax4: $P \varphi \wedge (\varphi \supset_N \psi) \supset P \psi$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi \wedge P \psi \supset P (\varphi \cdot \psi)$ ]"
5   axiomatization where Ax2a: "[P  $\varphi \vee^e P \sim \varphi$ ]"
6   definition God ("G") where "G x  $\equiv \forall \varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \forall \psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall \varphi. \varphi$  Ess. x  $\supset \Box(\exists^e x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset \Box(\exists^e y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^e x. G x) \supset \Diamond \Box(\exists^e y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P  $\varphi \wedge (\varphi \supset_N \psi) \supset P \psi$ ]"
```

Ax4:

Properties that are necessarily implied by a positive property are positive.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

No model found!

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9   axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10  lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^e x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset$   $\Box(\exists^e y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^e x. G x)$   $\supset$   $\Box(\exists^e y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^e x. G x)$   $\supset$   $\Box(\exists^e y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^e x. G x)$   $\supset$   $\Diamond\Box(\exists^e y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^e x. G x)$   $\supset$   $\Box(\exists^e y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23  lemma True nitpick[satisfy] oops —<No model found>
```

Consistency check with Nitpick:
... no model found ...

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Inconsistency detected
by theorem provers

The inconsistency follows via the
“empty essence lemma”:

$$(\lambda x. \perp) \text{Ess. } x$$



Adapt the notion of essence?

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4   axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5   axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6   definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7   abbreviation PropertyInclusion ("_⊃N_") where "φ ⊃N ψ ≡ □(∀Ey. φ y ⊃ ψ y)"
8   definition Essence ("_Ess._") where "φ Ess. x ≡ ∀ψ. ψ x ⊃ (φ ⊃N ψ)"
9   axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10  lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃Ex. φ x)"
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x ⊃ □(∃Ey. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ (∃Ex. G x) ⊃ □(∃Ey. G y)]" using Th2 by blast
18    have 2: "[ ◇(∃Ex. G x) ⊃ ◇□(∃Ey. G y)]" using 1 by blast
19    have 3: "[ ◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23  lemma True nitpick[satisfy] oops —<No model found>
24  lemma EmptyEssL: "[ (λy. ⊥) Ess. x]" using Essence_def by auto
```

Fig. 6 Gödel’s axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Inconsistency detected
by theorem provers

The inconsistency follows via the
“empty essence lemma”:

$(\lambda x. \perp) \text{Ess. } x$



Adapt the notion of essence?

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi. P \varphi \supset \varphi$  x"
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi \text{Ess. } x \equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall\varphi. \varphi \text{Ess. } x \supset \Box(\exists^e x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset \Box(\exists^e y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^e x. G x) \supset \Diamond\Box(\exists^e y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P  $\varphi \wedge (\varphi \supset_N \psi) \supset P \psi$ ]"
23  lemma True nitpick[satisfy] oops —<No model found>
24  lemma EmptyEssL: "[ $(\lambda y. \perp) \text{Ess. } x$ ]" using Essence_def by auto
25  theorem Inconsistency: False sledgehammer(Ax2a Ax3 Ax4 EmptyEssL NecExist_def) —<Proof found>
26  proof -
27    have 1: "[ $\neg(P (\lambda x. \perp))$ ]" using Ax2a Ax4 by blast
28    have 2: "[P ( $\lambda x. (\lambda y. \perp) \text{Ess. } x \supset \Box(\exists^e z. (\lambda y. \perp) z)$ )]" using Ax3 Ax4 NecExist_def by smt
29    have 3: "[P ( $\lambda x. \Box(\exists^e z. (\lambda x. \perp) z)$ )]" using 2 EmptyEssL by simp
30    have 4: "[P ( $\lambda x. \Box \perp$ )]" using 3 by auto
31    have 5: "[P ( $\lambda x. \perp$ )]" using 4 Ax2a Ax4 by smt
32    have 6: "[ $\perp$ ]" using 1 5 by blast
33    thus ?thesis by blast
34  qed
35 end
```

Inconsistency
... proved ...

Fig. 6 Gödel’s axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1 — Results

(Positive) properties as intensional predicates

‘ φ necessarily implies ψ ’ defined as $\Box \forall^E x . \varphi x \supset \psi x$

Ax1: The conjunction of two positive properties is positive.

Ax2a: Either a property or its complement is positive.

G(od-like): x is God-like iff it possesses all positive properties.

Ess(ence): φ is an essence of x iff all properties of x are necessarily implied by φ .

Ax2b: Positive properties are necessarily positive.

Theorem 1: Being God-like is the essence of every God-like entity.

E(xistence): x has existence iff all essences of x are necessarily exemplified.

Ax3: E(xistence) is a positive property.

Theorem 2: If x is God-like, then God-likeness is necessarily exemplified.

Theorem 3: The possible existence of a God-like entity implies the necessary existence of a God-like entity.

Ax4: Properties that are necessarily implied by a positive property are positive.

Found by automated theorem provers
(2016) — missed by the earlier
commentary

These postulates are **inconsistent**
(already in **modal logic K**)

Formalisation of Gödel's manuscript of 1970

(adapted notion of essence and generalized axiom Ax1Gen: Gödel2)

Ontologischer Beweis Feb. 10, 1970

$P(\varphi)$ φ is positive ($\varphi \in P$)

Ax. 1 $P(\varphi) \cdot P(\psi) \supset P(\varphi \cdot \psi)$ Ax. 2 $P(\varphi) \cdot \neg P(\psi) \supset P(\varphi \cdot \neg \psi)$

[1] $G(x) \equiv (\varphi) [P(\varphi) \supset \varphi(x)]$ (Gödel)

[2] $\varphi \text{ Ess. } x \equiv (\psi) [\psi(x) \supset N(\psi) [\varphi(y) \supset \psi(y)]]$ (Essence of x)

$p \supset_N q = N(p \supset q)$ Necessity

Ax. 2 $P(\varphi) \supset N P(\varphi)$
 $\neg P(\varphi) \supset N \neg P(\varphi)$ } because it follows from the nature of the property

Th. $G(x) \supset G \text{ Ess. } x$

Def. $E(x) \equiv (\varphi) [\varphi \text{ Ess. } x \supset N \exists x \varphi(x)]$ necessary Existence

Ax. 3 $P(E)$

Th. $G(x) \supset N(\exists y) G(y)$
hence $(\exists x) G(x) \supset N(\exists y) G(y)$
" $M(\exists x) G(x) \supset M N(\exists y) G(y)$ $M = possibility$
" $\supset N(\exists y) G(y)$

any two instances of x are nec. equivalent
exclusive or * and for any number of humanoids

$M(\exists x) G(x)$ means ^{the system of} all pos. props. as com-
patible This is true because of:

Ax. 4: $P(\varphi) \cdot \varphi \supset_N \psi : \supset P(\psi)$ which impl.

~~True~~ $\begin{cases} x=x & \text{is positive} \\ x \neq x & \text{is negative} \end{cases}$

But if a system S of pos. props. were incomp. it would mean that the sum prop. S (which is positive) would be $x \neq x$

Positive means positive in the moral aesth. sense (independently of the accidental structure of the world). Only when the act. true. $\supset$ it also means "attribution" as opposed to "privation" (or containing privation). This interprets the other part

if φ privation $\neg(x) N \neg \varphi(x)$ - otherwise $\varphi(x) \supset_N x \neq x$
hence $x \neq x$ positive or $x=x$ neg. satisfying Ax. 4 or the equiv. of pos. $M(\exists x) G(x)$

day

x i.e. the normal form in terms of elem. prop. contains a member without negation.

— proof in KB —

$\varphi \supset \Box \Diamond \varphi$

Formalisation of Gödel's manuscript of 1970 (adapted notion of essence and generalized axiom Ax1Gen: Gödel2)

— proof in KB —
 $\varphi \supset \Box \Diamond \varphi$

```

1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee$  P  $\sim$  $\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall \varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion (" $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall y. \varphi y \supset \psi y)$ "
8 (**) definition Essence ("Ess.") where " $\varphi$  Ess. x  $\equiv$   $\varphi$  x  $\wedge$   $\forall \psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall \varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists x. G x)$   $\supset$   $\Box(\exists y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists x. G x)$   $\supset$   $\Box(\exists y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists x. G x)$   $\supset$   $\Diamond\Box(\exists y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists x. G x)$   $\supset$   $\Box(\exists y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall \varphi$ .  $\Phi \varphi \supset$  P  $\varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall z. \varphi z \leftrightarrow (\forall \psi. \Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ $\Diamond(\exists x. G x)$ ]" using Ax2a Ax4 L by blast
29 theorem Th5: "[ $\Box(\exists x. G x)$ ]" using Th3 Th4 by blast
30 lemma MC: "[ $\varphi \supset \Box \varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: "[ $\forall x. (G x w \longrightarrow (\forall z. z x \supset \Box(\forall z. G z \supset z z)) w)$ ]" using Ax2a Ax2b God_def by smt
33   have 2: "[ $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall z. G z \supset Q)) w)$ ]" using 1 by force
34   have 3: "[ $(Q \supset \Box Q) w$ ]" using 2 Th5 Rsymm by blast
35   thus ?thesis by auto
36 qed
37 lemma PosProps: "[P ( $\lambda x. \top$ )  $\wedge$  P ( $\lambda x. x = x$ )]" using Ax2a Ax4 by blast
38 lemma NegProps: "[ $\neg$ P ( $\lambda x. \bot$ )  $\wedge$   $\neg$ P ( $\lambda x. x \neq x$ )]" using Ax2a Ax4 by blast
39 lemma UniqueEss1: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\forall y. \varphi y \leftrightarrow \psi y)$ ]" using Essence_def by smt
40 lemma UniqueEss2: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\varphi \equiv \psi)$ ]" nitpick[card i=1] oops —<Countermodel found>
41 lemma UniqueEss3: "[ $\varphi$  Ess. x  $\supset$   $\Box(\forall y. \varphi y \supset y \equiv x)$ ]" using Essence_def MC by auto
42 lemma Monotheism: "[G x  $\wedge$  G y  $\supset$  x  $\equiv$  y]" using Ax2a God_def by smt
43 lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44 lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
46 end

```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2 — Results

Essence repair due to Dana Scott

‘ φ necessarily implies ψ ’ iff $\Box \forall^E x . \varphi x \supset \psi x$

Ax1Gen: The conjunction **of arbitrarily many** positive properties is positive.

Ax2a: Either a property or its complement is positive.

G(od-like): x is God-like iff it possesses all positive properties.

Ess(ence): φ is an essence of x iff all properties of x are necessarily implied by φ **and x has property φ .**

Ax2b: Positive properties are necessarily positive.

Theorem 1: Being God-like is the essence of every God-like entity.

E(xistence): x has existence iff all essences of x are necessarily exemplified.

Ax3: E(xistence) is a positive property.

Theorem 2: If x is God-like, then God-likeness is necessarily exemplified.

Theorem 3: The possible existence of a God-like entity implies the necessary existence of a God-like entity.

Ax4: Properties that are necessarily implied by a positive property are positive.

Lemma L: God-likeness is a positive property.

Theorem 4: A God-like entity possibly exists.

Theorem 5: A God-like entity necessarily exists.

Generalization of Ax1: **Ax1Gen**

• and for any number of minimally

Axiom Ax1Gen: $(PosProps \Phi \wedge ConjOfPropsFrom \varphi \Phi) \supset P \varphi$

abbreviation "PosProps $\Phi \equiv \forall \varphi. \Phi \varphi \supset P \varphi$ "

abbreviation "ConjOfPropsFrom $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall \psi. \Phi \psi \supset \psi z))$ "

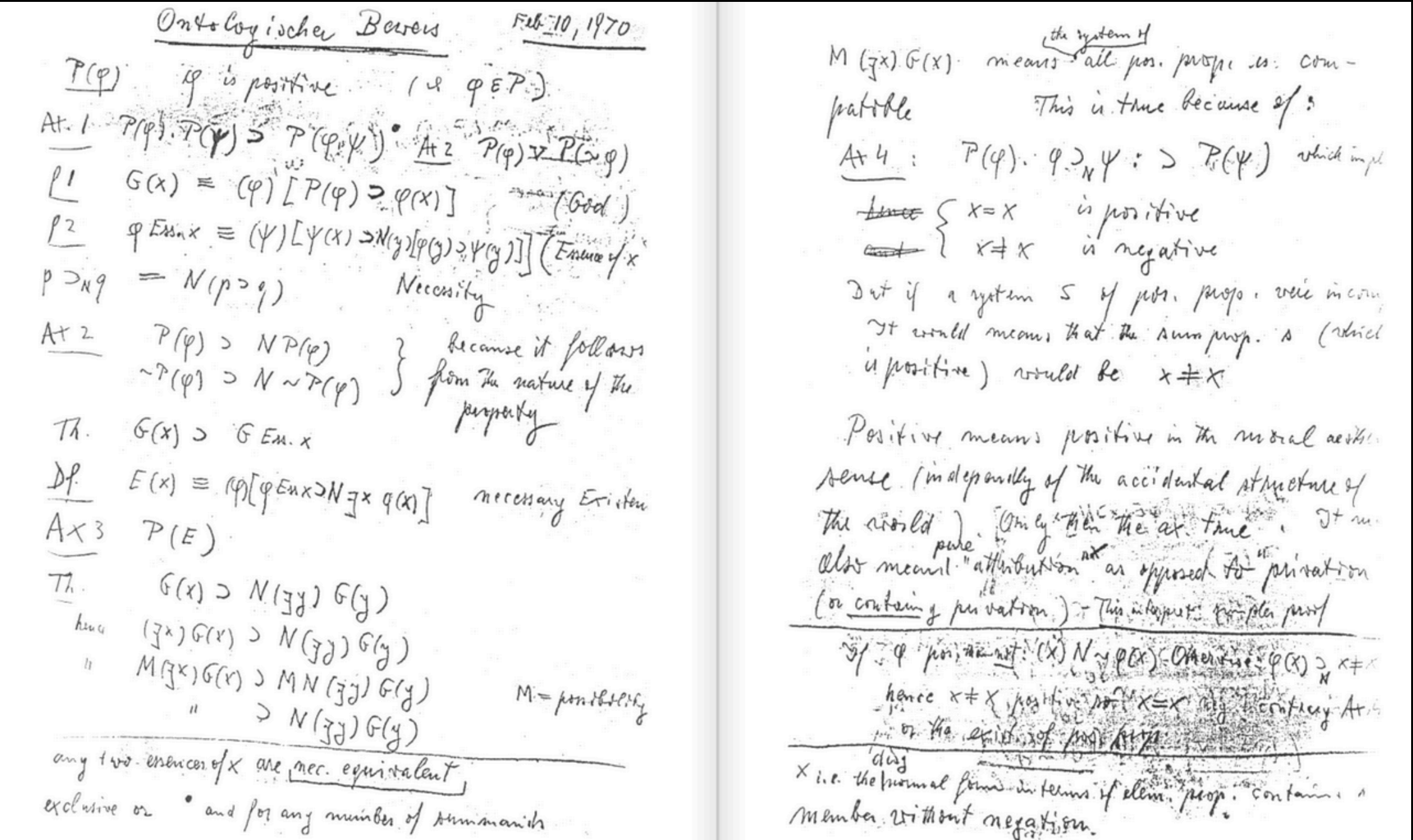
axiomatization where Ax1Gen: " $[(PosProps \Phi \wedge ConjOfPropsFrom \varphi \Phi) \supset P \varphi]$ "

Ax1Gen expressible in third-order logic

- Consistent
- Valid already in modal logic **KB**
- **Modal collapse** follows
- **Monotheism** follows
- **P** is an **ultrafilter**
- Self-identity: positive property
- Self-difference: non-positive prop.

Formalisation of Gödel's manuscript of 1970

(adapted property inclusion and generalized axiom Ax1Gen: Gödel3)



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— proof in KB —
 $\varphi \supset \Box \Diamond \varphi$

Note: This is a new variant

Formalisation of Gödel's manuscript of 1970

(adapted property inclusion and generalized axiom Ax1Gen: Gödel3)

— proof in KB —

$$\varphi \supset \Box \Diamond \varphi$$

```

1 theory GoedelVariantHOML3 imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4   axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5   axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6   definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7   (**) abbreviation PropertyInclusion ("⊃N") where "φ ⊃N ψ ≡ □ φ ≠ (λx. ⊥) ∧ ∀Ey. φ y ⊃ ψ y)"
8   definition Essence ("_Ess._") where "φ Ess. x ≡ ∀ψ. ψ x ⊃ (φ ⊃N ψ)"
9   axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10  lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃Ex. φ x)"
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x ⊃ □(∃Ey. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using Th2 by blast
18    have 2: "[◇(∃Ex. G x) ⊃ ◇□(∃Ey. G y)]" using 1 by blast
19    have 3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23  lemma True nitpick[satisfy,card=1,eval="|P (λx. ⊥)|"] oops —<Two models found of cardinality one>
24  abbreviation "PosProps Φ ≡ ∀φ. Φ φ ⊃ P φ"
25  abbreviation "ConjOfPropsFrom φ Φ ≡ □(∀Ez. φ z ↔ (∀ψ. Φ ψ ⊃ ψ z))"
26  axiomatization where Ax1Gen: "[ (PosProps Φ ∧ ConjOfPropsFrom φ Φ) ⊃ P φ ]"
27  lemma L: "[P G]" using Ax1Gen God_def by (smt (verit))
28  theorem Th4: "[◇(∃Ex. G x)]" sledgehammer[timeout=200](Ax2a L Ax1Gen) sorry —<Proof found>
29  theorem Th5: "[□(∃Ex. G x)]" using Th3 Th4 by blast
30  lemma MC: "[φ ⊃ □φ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31  proof - {fix w fix Q
32    have 1: "∀x.(G x w → (∀z. z x ⊃ □(∀Ez. G z ⊃ z z)) w)" using Ax2a Ax2b God_def by smt
33    have 2: "(∃x. G x w)→((Q ⊃ □(∀Ez. G z ⊃ Q)) w)" using 1 by force
34    have 3: "(Q ⊃ □Q) w" using 2 Th5 Rsymm by blast}
35  thus ?thesis by auto
36  qed
37  lemma PosProps: "[P (λx. T) ∧ P(λx. x = x)]" using Ax2a Ax4 L Th4 by smt
38  lemma NegProps: "[¬P(λx. ⊥) ∧ ¬P(λx. x ≠ x)]" using Ax2a Ax4 L Th4 by smt
39  lemma UniqueEss1: "[φ Ess. x ∧ ψ Ess. x ⊃ □(∀Ey. φ y ↔ ψ y)]" oops —<Unclear, open question>
40  lemma UniqueEss2: "[φ Ess. x ∧ ψ Ess. x ⊃ □(φ ≡ ψ)]" oops —<Unclear, open question>
41  lemma UniqueEss3: "[φ Ess. x ⊃ □(∀Ey. φ y ⊃ y ≡ x)]" using Essence_def MC by auto
42  lemma Monotheism: "[G x ∧ G y ⊃ x ≡ y]" using Ax2a God_def by smt
43  lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44  lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45  lemma True nitpick[satisfy,card=1,eval="|P (λx. T)|"] oops —<One model found of cardinality one>
46 end

```

Fig. 8 After an appropriate modification of the notion of necessary property inclusion in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 3 — Results

“Devil = ... (empty set)”

(Max III, p. 47)

“Remark: The empty set (the empty space) is an amphibian between nothing and something.”

(Max III, p. 85)

‘ φ necessarily implies ψ ’ iff $\Box \forall^E x . \varphi \neq \lambda x . \perp \wedge \varphi x \supset \psi x$



The empty set now no longer implies any other properties!

Ax1 Gen: The conjunction **of arbitrarily many** positive properties is positive.

Ax2a: Either a property or its complement is positive.

G(od-like): x is God-like iff it possesses all positive properties.

Ess(ence): φ is an essence of x iff all properties of x are necessarily implied by φ .

Ax2b: Positive properties are necessarily positive.

Theorem 1: Being God-like is the essence of every God-like entity.

E(xistence): x has existence iff all essences of x are necessarily exemplified.

Ax3: E(xistence) is a positive property.

Theorem 2: If x is God-like, then God-likeness is necessarily exemplified.

Theorem 3: The possible existence of a God-like entity implies the necessary existence of a God-like entity.

Ax4: Properties that are necessarily implied by a positive property are positive.

Lemma L: God-likeness is a positive property.

Theorem 4: A God-like entity possibly exists.

Theorem 5: A God-like entity necessarily exists.

- Consistent
- Valid already in modal logic **KB**
- **Modal collapse** follows
- **Monotheism** follows
- **P** is an **ultrafilter**
- Self-identity: positive property
- Self-difference: non-positive prop.

Gödel 2

Summary

requires the generalized axiom **Ax1Gen**

- modified: **essence**
- no existential import
- consistent
- possible and necessary existence of G
- verified already in logic **KB**
- derived results:
 - **modal collapse**
 - **monotheism**
 - **P is an ultrafilter**

Gödel 3

Summary

requires the generalized axiom **Ax1Gen**

- modified: property **inclusion**
- no existential import
- consistent
- possible and necessary existence of G
- verified already in logic **KB**
- derived results:
 - **modal collapse**
 - **monotheism**
 - **P is an ultrafilter**

Changing the quantifiers for entities from
'varying-domain' to 'constant-domain'? —> results reconfirmed

Gödel 2

Summary

requires the generalized axiom **Ax1Gen** (≥ 2)

- modified: **essence**
- no existential import
- consistent
- possible and necessary existence of G
- verified already in logic **KB**
- derived results:
 - **modal collapse**
 - **monotheism**
 - **P is an ultrafilter**

Gödel 3

Summary

requires the generalized axiom **Ax1Gen** (≥ 2)

- modified: property **inclusion** (**only for Ess**)
- no existential import
- consistent
- possible and necessary existence of G
- verified already in logic **KB**
- derived results:
 - **modal collapse**
 - **monotheism**
 - **P is an ultrafilter**

Changing the quantifiers for entities from
'varying-domain' to 'constant-domain'? $\rightarrow$ results reconfirmed

And the Devil?

“Namely, in order to recognise a thing {a}, I must have: 1.) certain fundamental assignments of certain empirical things (perhaps beginning with God = universal set*; Devil = an irrefutably false system* (or empty set); ...” (Max II, p. 47)

Rem<ark> (Theol<ogy>): The perfectly clear recognition of the {perfectly} evil one — that he can never achieve his aim — means his death. (Max IX, p. 36)

And the Devil? — Variant 1

(Credits to Chad Brown)

x is a Devil iff
x possesses all
non-positive
properties.

Axioms as before



Necessarily, no Devil
exists.

```
1 theory ThereIsNoEvil1 imports GoedelVariantHOML2
2 begin
3 definition Evil ("Evil") where "Evil x  $\equiv \forall \varphi. \neg P \varphi \supset \varphi x$ "
4 theorem NecNoEvil: "[ $\Box(\neg(\exists^E x. \text{Evil } x))$ ]" sledgehammer(Ax2a Ax4 Evil_def) —<Proof found>
5 proof -
6   have "[ $\neg P(\lambda y. \bot)$ ]" using Ax2a Ax4 by blast
7   hence "[ $(\forall^E x. \text{Evil } x \supset (\lambda y. \bot) x)$ ]" using Evil_def by auto
8   hence "[ $(\forall^E x. \text{Evil } x \supset \bot)$ ]" by auto
9   hence "[ $(\exists^E x. \text{Evil } x) \supset \bot$ ]" by auto
10  hence "[ $\neg(\exists^E x. \text{Evil } x)$ ]" by blast
11  thus ?thesis by blast
12 qed
13 end
```

Gödel 2

Fig. 10 Importing Gödel's modified axioms from Fig. 7 we can prove that necessarily there exists no entity that possesses all non-positive (=negative) properties

```
1 theory ThereIsNoEvil2 imports GoedelVariantHOML3
2 begin
3 definition Evil ("Evil") where "Evil x  $\equiv \forall \varphi. \neg P \varphi \supset \varphi x$ "
4 theorem NecNoEvil: "[ $\Box(\neg(\exists^E x. \text{Evil } x))$ ]" sledgehammer(Ax1Gen Ax2a Evil_def) oops —<Proof found>
5 end
```

Gödel 3

Fig. 11 Importing Gödel's modified axioms from Fig. 8 we can prove that necessarily there exists no entity that possesses all non-positive (=negative) properties

Also confirmed for
Gödel 3

Relative to the axioms — whether this tells us anything about reality depends on whether one accepts them.

And the Devil? — Variant 2 (Scott)

```
1 theory EvilDerivable imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 definition Evil ("Evil") where "Evil x  $\equiv$   $\forall \varphi. \neg P \varphi \supset \varphi x$ "
5 definition Essence ("_Ess.") where " $\varphi$  Ess. x  $\equiv$   $\varphi x \wedge (\forall \psi. \psi x \supset \Box(\forall^E y. \varphi y \supset \psi y))$ "
6 definition NecExist ("E") where "E x  $\equiv$   $\forall \varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
7 axiomatization where A1: " $\neg P \varphi \leftrightarrow P \neg \varphi$ "
8 axiomatization where A2: " $\neg P \varphi \wedge \Box(\forall^E y. \varphi y \supset \psi y) \supset \neg P \psi$ "
9 axiomatization where A4: " $\neg P$  Evil"
10 axiomatization where A3: " $\neg P \varphi \supset \Box(\neg P \varphi)$ "
11 axiomatization where A5: " $\neg P E$ "
12 lemma True nitpick[satisfy,card i=1,eval="P ( $\lambda x. \bot$ )",eval="P ( $\lambda x. \top$ )"] oops —<Model found>
13 theorem T1: " $\neg P \varphi \supset \Diamond(\exists^E x. \varphi x)$ " using A1 A2 by blast
14 theorem T2: " $\Diamond(\exists^E x. \text{Evil } x)$ " using A4 T1 by blast
15 theorem T3: " $\text{Evil } x \supset \text{Evil } \text{Ess. } x$ " using A1 A3 Essence_def Evil_def by (smt (verit, best))
16 theorem T4: " $\Diamond(\exists^E x. \text{Evil } x) \supset \Box(\exists^E y. \text{Evil } y)$ " using A5 Evil_def NecExist_def Rsymm T3 by smt
17 theorem T5: " $\Box(\exists^E x. \text{Evil } x)$ " using T2 T4 by presburger
18 lemma MC: " $\varphi \supset \Box \varphi$ " sledgehammer(A1 A3 T5 Evil_def Rsymm) oops —<proof found>
19 lemma PosProps: " $P (\lambda x. \bot) \wedge P (\lambda x. x \neq x)$ " using A1 A2 by blast
20 lemma NegProps: " $\neg P (\lambda x. \top) \wedge \neg P (\lambda x. x = x)$ " using A1 A2 by blast
21 lemma UniqueEss1: " $\varphi$  Ess. x  $\wedge \psi$  Ess. x  $\supset \Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ " using Essence_def by smt
22 lemma UniqueEss2: " $\varphi$  Ess. x  $\wedge \psi$  Ess. x  $\supset \Box(\varphi = \psi)$ " nitpick[card i=2] oops —<Countermodel found>
23 lemma Monoevilism: " $\text{Evil } x \wedge \text{Evil } y \supset x \equiv y$ " using A1 Evil_def by smt
24 lemma Filter: " $\text{Filter } (\lambda \varphi. \neg P \varphi)$ " using A1 Evil_def Rsymm T1 T5 by (smt (verit, best))
25 lemma UltraFilter: " $\text{UFilter } (\lambda \varphi. \neg P \varphi)$ " using Filter A1 by blast
26 end
```

Fig. 20 The necessary existence of an Evil-like entity proved from (controversially) modified assumptions

x is a Devil iff x possesses all
non-positive properties

Mirroring of all axioms (Scott)
for non-positive properties



Necessarily, there exists
a Devil

But: The properties of this Devil are absolutely identical to the properties of
Gödel's God-like entity!

The audit currently continues



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[Submitted on 4 Aug 2026 (v1), last revised 26 Aug 2026 (this version, v3)]

A Comment on Modal Collapse and Ultrafilters in Gödel's Ontological Argument

Christoph Benz Müller

Gödel's ontological argument is modal by design. Possibility and necessity are among its central concerns, and connected interpretation decisions -- extensions or intensions for the positive properties, constant or varying quantifier domains, the choice of modal logic, etc. -- subtly shape the argument and, in particular, whether the modal collapse is implied or not. It is tempting to abstract the modality away and to read the positive properties as merely forming an ultrafilter -- which, however, may replace Gödel's argument with a perhaps more manageable alternative that may have little to do with Gödel's original intention. In this spirit, Odifreddi and Gomes argue that modal collapse is an intrinsic structural feature of any theory that characterises positive properties as an ultrafilter and identifies God as its principal generator. This comment shows, with machine-verified counterexamples, that the structural claim is false: what drives the collapse is primarily not the filter structure but the rigidity of positivity -- itself a thoroughly modal condition. Moreover, two claims of Odifreddi and Gomes are corrected: one on the derivability of Gödel's Theorem IV, one on the extensionality of positivity. All results are machine-checked in Isabelle/HOL.

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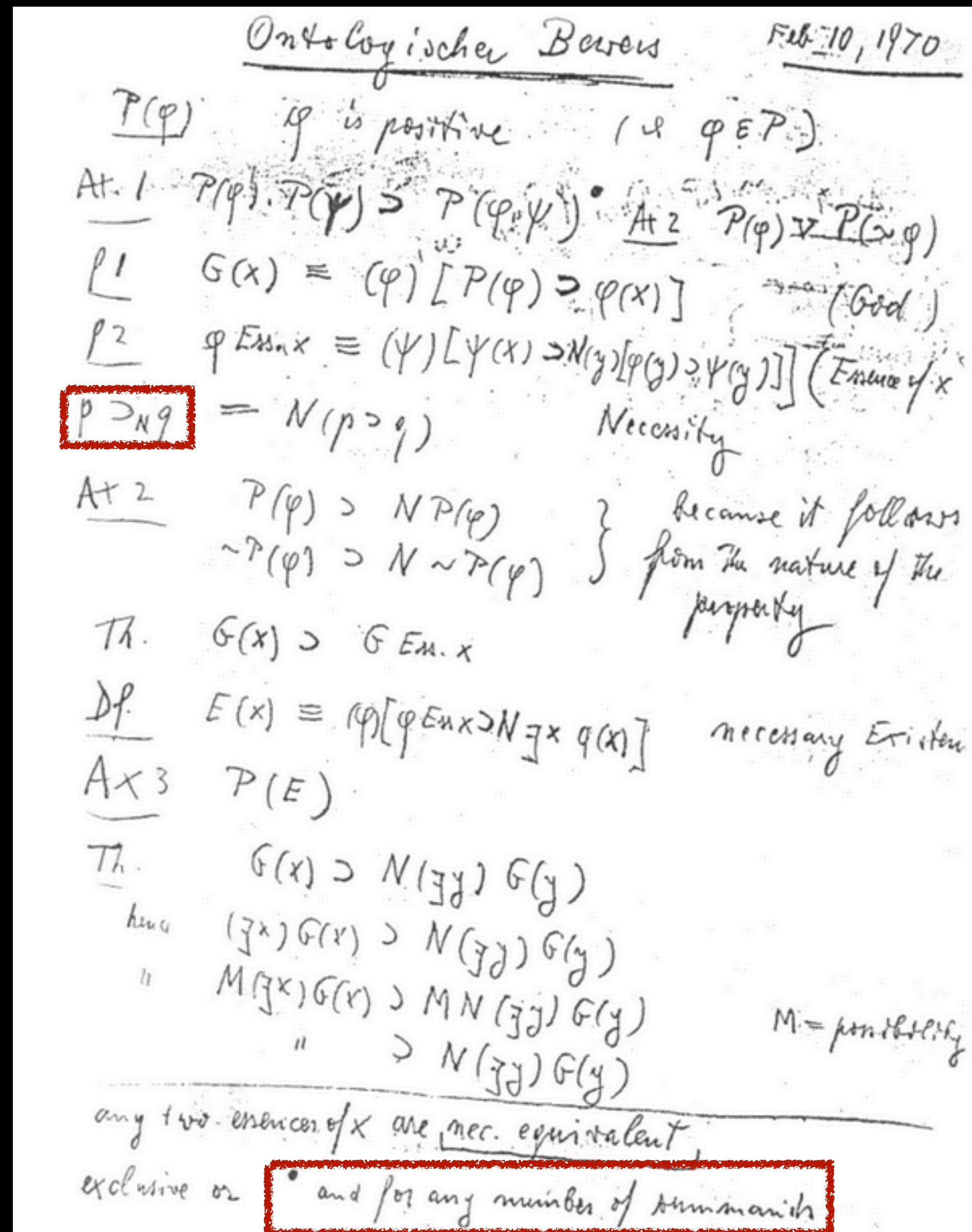
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Personal conclusion



Gödel's scriptum variant of 1970

- with modified property inclusion (**for Ess**)
- with the generalised Ax1Gen (**for ≥ 2**)

is my preferred 'reading' of his modal ontological argument.

I also share the view of e.g. Kovač and Mühlenbeck (and others) that Gödel did not merely accept the modal collapse, but intended it.

Investigating Gödel's "universal set" (maximal infinity) is fascinating further work (type vs. set theory).

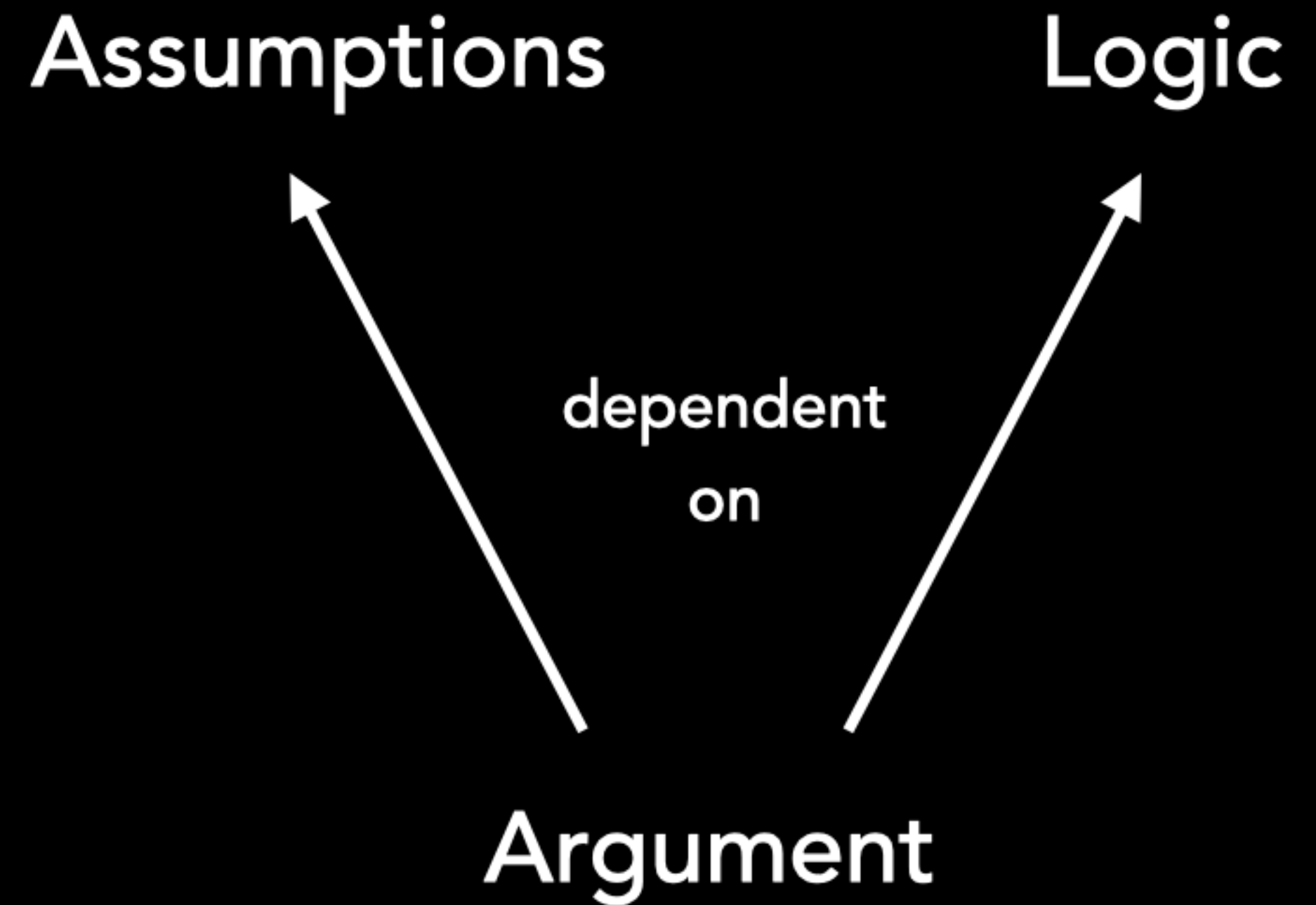
member without negation.

Important Clarification

No!

We haven't proven the
existence of God using a
computer!

We have formally analysed a
prominent argument for the
existence of God using a
computer.



Gödel's scriptum variant (1970) and mathematical realism

God = (maximally) infinite/unfathomable?

Logic and Logical Philosophy
Online First Articles (2026)
DOI: [10.12775/LLP.2026.007](https://doi.org/10.12775/LLP.2026.007)

Cordelia Mühlenbeck 

Christoph Benz Müller 

On the Maximality of Positive Properties and Modal Collapse in Variants of Gödel's Ontological Proof for the Existence of God

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Christoph Benz Müller, Daniel Kirchner, Luca Pasetto

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Gödel 2 — as a starting point ...

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter ChurchTTinHOML
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where
5 axiomatization where
6 definition God ("G") where
7 abbreviation PropertyInclusion (" $\supset_N$ ") where
8 definition Essence ("_Ess._") where
9 axiomatization where
10 lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]"
11 theorem Th1: "[ $G x \supset G \text{Ess. } x$ ]"
12 definition NecExist ("E") where
13 axiomatization where
14 theorem Th2: "[ $G x \supset \Box(\exists^E y. G y)$ ]"
15 theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]"
16 axiomatization where
17 lemma True nitpick[satisfy,card=1,eval="[ $P(\lambda x. \perp)$ ]" ] oops —<One model found of cardinality one>
18 abbreviation CollOfPosProps  $\Phi \equiv \forall \varphi. \Phi \varphi \supset P \varphi$ 
19 abbreviation ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall \psi. \Phi \psi \supset \psi z))$ 
20 axiomatization where
21 lemma L: "[ $P G$ ]"
22 theorem Th4: "[ $\Diamond(\exists^E x. G x)$ ]"
23 theorem Th5: "[ $\Box(\exists^E x. G x)$ ]"
24 lemma MC: "[ $\varphi \supset \Box \varphi$ ]" —<Proof found with sledgehammer>
25 proof - {fix w fix Q
26 have 1: "[ $\forall x. (G x w \longrightarrow (\forall Z. Z x \supset \Box(\forall^E z. G z \supset Z z)) w)$ ]" using Ax2a Ax2b God_def by smt
27 have 2: "[ $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^E z. G z \supset Q)) w)$ ]" using 1 by force
28 have 3: "[ $(Q \supset \Box Q) w$ ]" using 2 Th5 Rsymm by blast
29 thus ?thesis by auto qed
30 lemma PosProps: "[ $P(\lambda x. \top) \wedge P(\lambda x. x = x)$ ]" using Ax2a Ax4 by blast
31 lemma NegProps: "[ $\neg P(\lambda x. \perp) \wedge \neg P(\lambda x. x \neq x)$ ]" using Ax2a Ax4 by blast
32 lemma UniqueEss1: "[ $\varphi \text{Ess. } x \wedge \psi \text{Ess. } x \supset \Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ ]" using Essence_def by smt
33 lemma UniqueEss2: "[ $\varphi \text{Ess. } x \wedge \psi \text{Ess. } x \supset \Box(\varphi \equiv \psi)$ ]" nitpick[card i=1] oops —<Countermodel found>
34 lemma Monotheism: "[ $G x \wedge G y \supset x \equiv y$ ]" using Ax2a God_def by smt
35 lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
36 lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
37 lemma True nitpick[satisfy,user_axioms,card=1,eval="[ $P(\lambda x. \perp)$ ]" ] oops —<One model found of cardinality one>
```

Gödel 2 — plus mathematical objects

Assumption: The objects of mathematics ‘exist’ (e.g. natural numbers n , Peano axioms).

Predicates: $\lambda x . x = n$ and $\lambda x . x \neq n$

Implications for the cardinality of the positive properties P :

- P is infinite
- P is uncountably infinite

Claim (verification is further work):

if we again assume such properties to exist, then the generalized axiom Ax1Gen forces the cardinality of P into the maximally infinite

On the maximality of positive properties and modal collapse in variants of Gödel’s ontological proof of God

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On Absolute Infinity and Modal Collapse in Gödel’s Ontological Proof

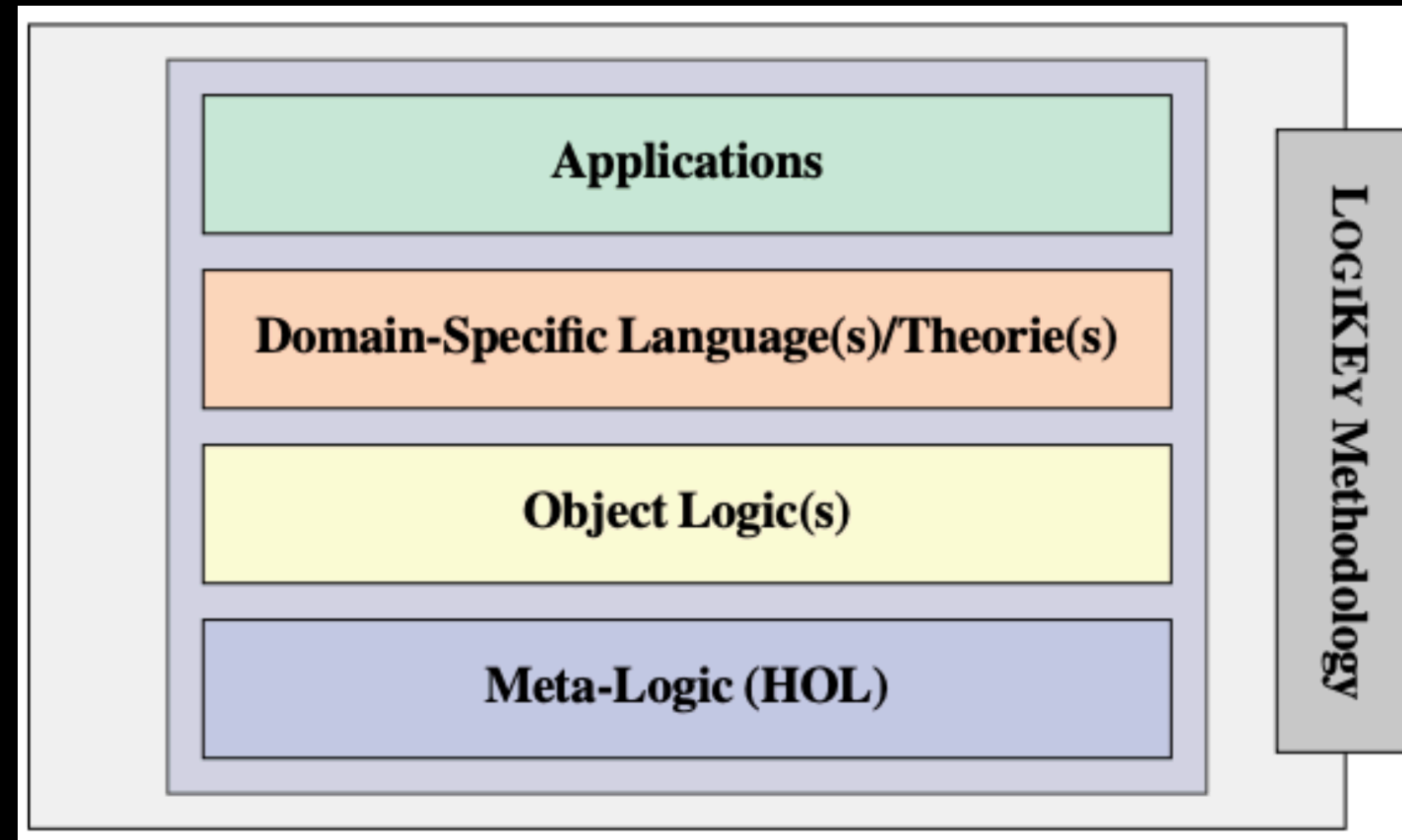
Cordelia Mühlenbeck, Christoph Benzmüller · *Logic, Knowledge, and Tradition: Essays in Honor of Srećko Kovač* · In print

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```
38 —<Assuming 2 different entities leads to 2 different positive properties, assuming 3 leads to 4, etc.>
39 lemma H: "[P (λe. T) ∧ (P (λe. e=a) ∨ P (λe. e≠a))]" using PosProps Ax2a by auto
40 lemma PP2: assumes "[∃(a::e) b. a ≠ b]" shows "[∃p q. p≠q ∧ P p ∧ P q]" by (smt (verit, del_insts) H assms)
41 lemma PP4: assumes "[∃(a::e) b c. a ≠ b ∧ a ≠ c ∧ b ≠ c]"
42   shows "[∃p q r s. p≠q ∧ p≠r ∧ p≠s ∧ q≠r ∧ q≠s ∧ r≠s ∧ P p ∧ P q ∧ P r ∧ P s]" by (smt (verit, del_insts) H assms)
43 —<Assuming infinitely many different entities leads to at least 2, 4, etc., many different positive properties>
44 lemma PP2': assumes "[Inf0]" shows "[∃p q. ((p≠q) ∧ P p ∧ P q)]" by (metis (lifting) Ax2a Inf0_def PosProps assms)
45 lemma PP4': assumes "[Inf0]" shows "[∃p q r s. p≠q ∧ p≠r ∧ p≠s ∧ q≠r ∧ q≠s ∧ r≠s ∧ P p ∧ P q ∧ P r ∧ P s]"
46 proof - have "[Inf0] ⇒ [∃p::e. ∃q::e. ∃r::e. p≠q ∧ p≠r ∧ q≠r]" unfolding Inf0_def apply simp by metis
47   thus ?thesis using PP4 assms by blast qed
48 —<Theorem MetaInf: infinitely many different entities lead to infinitely many positive properties: proof at meta-level.>
49   —<Various definitions and automatically proved lemmata are not shown here.> (*{{{ *}) [9 lines]
59 lemma MetaInf: "[Inf0] ⇒ infinite (UNIV::e set)" (*{{{ *}) [6 lines]
60 lemma InfPosSet_a: "infinite (UNIV::e set) ⇒ infinite (P w)" (*{{{ *}) [5 lines]
72 lemma InfPosSet_b: "[Inf0] ⇒ infinite (P w)" using InfPosSet_a MetaInf by blast (*{{{ *}) [6 lines]
79 theorem InfPosSet assumes "[Inf0]" shows "[¬(Finite P)]" by (metis Finite_def InfPosSet_b Nat_finite assms)
80 —<CantorArgument: Assuming infinitely many different entities leads to uncountably many different positive properties.>
81 definition S (f::nat⇒e) ≡ (λx::e. if x ∈ range f then f (Suc (inv f x)) else x)"
82 definition Enc (f::nat⇒e) s ≡ (λ(B::e⇒bool) (x::e) (u::i). x = f 0 ∨ (∃y. s y = x ∧ B y))"
83 definition D (f::nat⇒e) s (w::i) ≡ (λx::e. if F x (f 0) w then ¬ F x (s x) w else F x (s x) w)"
84 lemma Inf_inj: "[Inf0] ⇒ ∃f::nat⇒e. inj f" using MetaInf infinite_countable_subset by blast
85 lemma Enc0: "Enc f s B (f 0) w" using Enc_def by simp
86 lemma EncS: "inj f ⟶ (Enc f (S f) B (S f y) w ⟷ B y)"
87   by (smt Enc_def S_def inj_Suc inv_f_f inv_int_injective nat.distinct rangel)
88 lemma L1: "inj (f::nat⇒e) ⇒ (¬(∃F. ∀p. P p ⊃ (∃e::e. F e = p))) w" by (smt Enc0 EncS Ax2a D_def)
89 theorem CantorArgument: assumes "[Inf0]" shows "[¬(∃F. ∀p. P p ⊃ (∃e::e. F e = p))]" using assms Inf_inj L1 by metis
90 end
```

```
22 —<Formalized notions: equipollence, cardinality, different notion of infinity>
23 abbreviation Ex1B (binder "∃₁" [8]9) where "∃₁x. Φ(x) ≡ (∃x. Φ(x) ∧ (∀z. Φ(z) ⊃ (z = x)))"
24 definition Equipollent ≡ (λp::'a⇒σ. λq::'b⇒σ. (∃s. (∀x. p x ⊃ q (s x)) ∧ (∀y. q y ⊃ (∃₁x. p x ∧ (y = s x)))))"
25 definition Cardinality ≡ (λp::'b⇒σ. Equipollent p)"
26 definition CardinalNumbers ≡ (λu::('b⇒σ)⇒σ. (∃p::('b⇒σ). (u = (Equipollent p))))"
27 definition Inf0 ≡ ∃r::e⇒σ. ∀x y z. ((∃w. r x w) ∧ ¬(r x x) ∧ (r x y ⊃ (r y z ⊃ r x z)))"
28 definition Inf1 ≡ ∃n::e⇒σ. ∃m::e⇒σ. (∀x. m x ⊃ n x) ∧ (∃u. n u ∧ ¬(m u)) ∧ (Equipollent n m)"
29 definition Inf2 ≡ ∃n::e⇒σ. ∃u. ∃s. n u ∧ (∀x. n x ⊃ n (s x)) ∧ (∀x. n x ⊃ (s x)≠u) ∧ (∀x y. (n x ∧ n y ∧ s x=s y) ⊃ x=y)"
30 —<Definitions/abbreviations for modalised natural numbers (modeled as analysable objects naturally contained in HOL)>
31 named theorems Nat
32 definition Zero ("0") where [Nat]: "0 ≡ ((=) (λx. ⊥))"
33 definition Succ ("S") where [Nat]: "S ≡ (λn. λp. (∃x. p x ∧ (n (λt. (t ≠ x) ∧ p t))))"
34 definition Naturals ("N") where [Nat]: "N ≡ (λn. (∀p. (p 0 ∧ (∀x. p x ⊃ p (S x))) ⊃ p n))"
35 definition Finite ("Finite") where [Nat]: "Finite ≡ (λp. (∃n. N n ∧ n p))"
```

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Logical Pluralism with LogiKey

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